

Global Overlap Dashboard

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Purpose

This document defines the four Key Performance Indicators (KPIs) displayed on the Global Overlap Dashboard. Each KPI is a single, meaningful number that directly supports the project's central analytical question:

"Which globally trending artists and songs are missing from local markets, and how do these patterns change over time?"

The dashboard is designed for data-curious users. It does not require interaction — it presents a data-driven argument that the Discovery Gap is real, measurable, and consistent across regions and time. The KPIs are the headline of that argument, displayed as prominent summary figures at the top of the dashboard before the user engages with any trend charts or breakdowns.

Important Context — Historical Data

This is a historical analysis, not a real-time prediction tool. The dataset covers two periods:

- Dataset 1: January 2017 – December 2021
- Dataset 2: October 2023 – June 2025
- Known gap: December 2021 – October 2023 (22 months, no data)

All KPIs reflect patterns observed across this historical dataset. The value of these metrics is not in predicting what will trend next — it is in demonstrating how music crosses borders, and how large the gap between global momentum and local discovery consistently is. That is the analytical story this dashboard tells.

KPI Overview

KPI
1

Global Overlap Rate

Percentage of charting songs that appeared in 2+ countries

The headline number. Proves the silos are real and measurable.

KPI
2

Average Discovery Gap

Average days between a song's first chart appearance and its second country

Makes the abstract problem concrete. Quantifies the literal lag in cross-regional discovery.

KPI
3

Most Globally Isolated Region

Country with the lowest overlap with global chart trends

Gives the geographic dimension a headline. Identifies where the Discovery Gap is largest.

KPI
4

Peak Cross-Regional Reach

Max countries a single song simultaneously charted in

The ceiling number. Makes the gap meaningful by showing what maximum global penetration looks like.

Detailed KPI Definitions

KPI 1 — Global Overlap Rate

What it is: Of all songs that charted across any country during the dataset period, what percentage appeared in the charts of more than one country?

Why it matters: This is the headline number. A low overlap rate is direct evidence that music discovery is regionally siloed — most charting songs stay local. Tracking how this number changes over time tells the trend story of whether markets are becoming more or less connected.

How it is calculated:

```
Global Overlap Rate (%) =  
(Songs that charted in 2+ countries / Total unique charting songs) * 100
```

Display: Percentage. Bold headline KPI with a supporting trend indicator (e.g. up/down vs. prior period).

SP Inputs: Date range (defaults to full dataset).

SP Outputs: Overlap percentage, total unique songs, songs charting in 2+ countries.

Notes: Calculated separately for each dataset period due to the data gap. The two periods should not be averaged together without clearly labeling them as distinct timeframes.

KPI 2 — Average Discovery Gap (Days)

What it is: When a song first appears on any country's chart, how many days pass on average before it appears on a second country's chart?

Why it matters: This is the most compelling KPI because it makes the abstract problem concrete and measurable. It quantifies the literal lag — the time between a song gaining momentum somewhere in the world and that momentum crossing a border. A large average gap is direct evidence that chart algorithms and regional listening habits slow cross-regional discovery even when the music is globally accessible.

How it is calculated:

```
For each song:  
  First chart date = MIN(snapshot_date) across all countries  
  Second country date = MIN(snapshot_date) WHERE country != first country  
  
Discovery Gap (days) = Second country date - First chart date  
  
Average Discovery Gap = AVG(Discovery Gap)  
  across all songs that charted in 2+ countries
```

Display: Number of days. Bold headline KPI (e.g. "42 days on average").

SP Inputs: Date range, minimum country count threshold (default: 2).

SP Outputs: Average gap in days, median gap in days, sample size (songs included in calculation).

Notes: Songs that only ever charted in one country are excluded — they have no second country date. The median should be surfaced alongside the average, as chart data can have extreme outliers that skew the mean significantly.

KPI 3 — Most Globally Isolated Region

What it is: Which country or region has the lowest overlap with global chart trends — meaning the smallest percentage of its charting songs also appear in other countries' charts?

Why it matters: This gives the geographic dimension a headline number. It identifies where the Discovery Gap is largest and makes the regional variation in music culture tangible. It also gives data-curious users an immediate point of interest to explore further on the globe.

How it is calculated:

```
For each country:  
  Local songs = COUNT(DISTINCT song_id) WHERE country = @country  
  Shared songs = COUNT(DISTINCT song_id) WHERE country = @country  
    AND song_id IN (songs charting in any other country)  
  
Isolation Score (%) =  
  ((Local songs - Shared songs) / Local songs) * 100  
  
Most Isolated = Country with highest Isolation Score
```

Display: Country name + isolation score percentage (e.g. "Indonesia — 84% locally unique").

SP Inputs: Date range.

SP Outputs: Ranked list of all countries by isolation score. Dashboard displays only the top result as a KPI; the full ranked list feeds a secondary chart component.

Notes: Countries with very few total chart entries may produce misleadingly high isolation scores. Apply a minimum chart entry threshold (suggested: 100+ entries) before including a country in this calculation.

KPI 4 — Peak Cross-Regional Reach

What it is: The highest number of countries a single song simultaneously charted in at any point in the dataset.

Why it matters: This is the ceiling number — what maximum global penetration actually looks like. It makes the Discovery Gap more meaningful by contrast. If the most globally connected song charted in 58 countries simultaneously and the average song charts in 3, that gap is the story.

How it is calculated:

```
For each song, for each snapshot_date:
  Country count = COUNT(DISTINCT country_id)
                  WHERE song_id = @song
                  AND snapshot_date = @date

Peak Reach = MAX(Country count) across all songs and all dates
```

Display: Number of countries + song title + artist name (e.g. "58 countries — 'Blinding Lights' by The Weeknd").

SP Inputs: Date range.

SP Outputs: Peak country count, song_id, song title, artist name, date of peak reach.

Notes: This is the one KPI that surfaces a specific song and artist by name, making it the most human of the four. The song title and artist name should be displayed as a Vinyl card or similar on-brand component rather than plain text.

Summary Table

KPI	What It Measures	Output	SP Inputs
Global Overlap Rate	% of songs charting in 2+ countries	Percentage	Date range
Average Discovery Gap	Days between first and second country chart appearance	Days (integer)	Date range, min threshold
Most Globally Isolated Region	Country with lowest overlap with global trends	Country + %	Date range
Peak Cross-Regional Reach	Max countries a single song charted in simultaneously	Count + song/artist	Date range

